



ENEL301

Fundamentals of Engineering
Economics and Management

Sustainability

Outline of sustainability sessions

Monday 2 October 08:00 – 09:00	Case Study, Bougainville Copper Limited
Wednesday 2 October 15:00 – 16:00	Triple Bottom Line
Friday 4 October 12:00 – 13:00	Legislation and environmental management
Monday 7 October 08:00 – 09:00 15:00 – 16:00	Corporate sustainability reporting Greenhouse gas reporting Triple Bottom Line (Virtual Whiteboard)
Wednesday 9 October 15:00 – 16:00	Life cycle assessment #1
Friday 11 October 12:00 – 13:00	Life cycle assessment #2
Monday 14 October 08:00 – 09:00 15:00 – 16:00	Life cycle assessment #3 Life Cycle Assessment (Virtual Whiteboard)

- Students will have a framework for analysis of sustainability and exposure to current sustainability issues and their cultural implications

Washington Accord

Environment and Sustainability	Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.
The Engineer and Society	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

What does an engineer do?

“Engineers are people with specialised technical knowledge who conceive, deliver, operate and sustain artificial objects, systems and processes. These enable people to do more with less...effort, time, materials, energy, uncertainty, health risk and environmental disturbances.” – James Trevelyan

→ The modern engineer needs to reconnect the application of their specialist human skills within the rules of our larger global **context**.

When engineers stuff it up....

VW: Falsified emissions claims



Rio Tinto: Blowing up human history



Also a little closer to home...

Air New Zealand:
Helping the Saudi Navy



<https://www.1news.co.nz/2021/03/16/air-nz-not-in-breach-of-export-rules-after-helping-saudi-navy/#:~:text=Officials%20have%20decided%20Air%20New%20Zealand%20will%20not,was%20not%20in%20breach%20of%20export%20control%20orders.>

Rio Tinto: Ooops, yet another one

Rio Tinto apologises for
handling of Southland waste, in
Tiwai clean up commitment

27 May, 2021 05:40 PM

© 4 minutes to read



<https://www.nzherald.co.nz/business/rio-tinto-apologises-for-handling-of-southland-waste-in-tiwai-clean-up-commitment/L37LNAEZ2T6HQ7IJDUMOKK7F6A/>

A professional imperative



Engineering New Zealand takes ethics and professionalism very seriously. We hold our engineers accountable for their work on behalf of the New Zealand public. Our ethics are based on professional competence, personal integrity and social responsibility. Members of Engineering New Zealand are subject to the Code and commit to it each year.

Our Code of Ethical Conduct outlines the high standards of ethical behaviour expected of all members of Engineering New Zealand.

1. Take reasonable steps to safeguard health and safety
2. Have regard to effects on environment
3. Report adverse consequences
4. Act competently
5. Behave appropriately
6. Inform others of consequences of not following advice
7. Maintain confidentiality
8. Report breach of Code

- Potential law and regulations changes coming, meaning that you may be automatically be registered as an engineer upon or after graduation!
- You may have to abide by such a code, even though you might not know it...

A professional imperative

CODE OF ETHICAL CONDUCT

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1. Take reasonable steps to safeguard health and safety

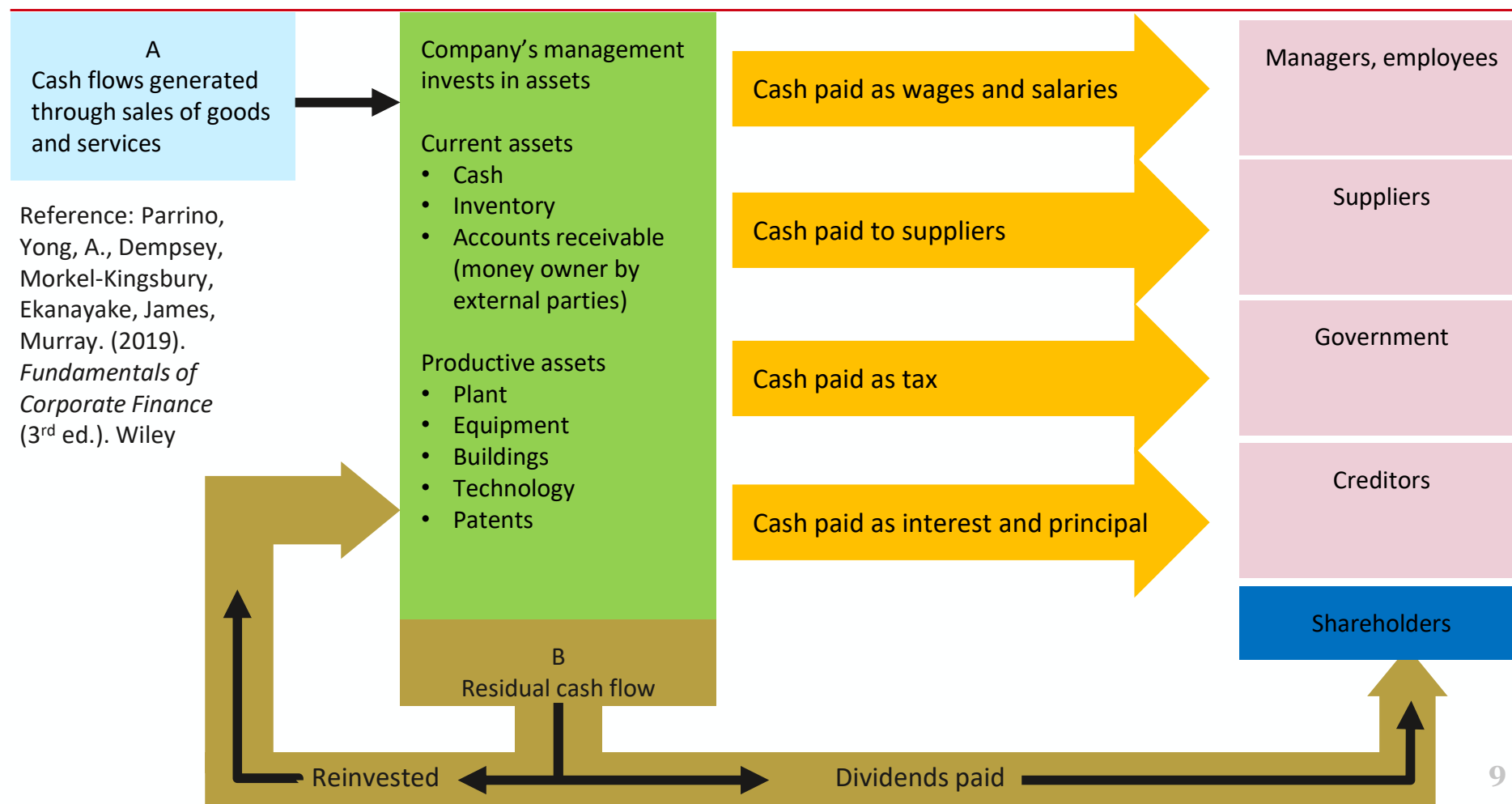
You must, in the course of your engineering activities, take reasonable steps to safeguard the health and safety of people.

2. Have regard to effects on environment

You must, in the course of your engineering activities,

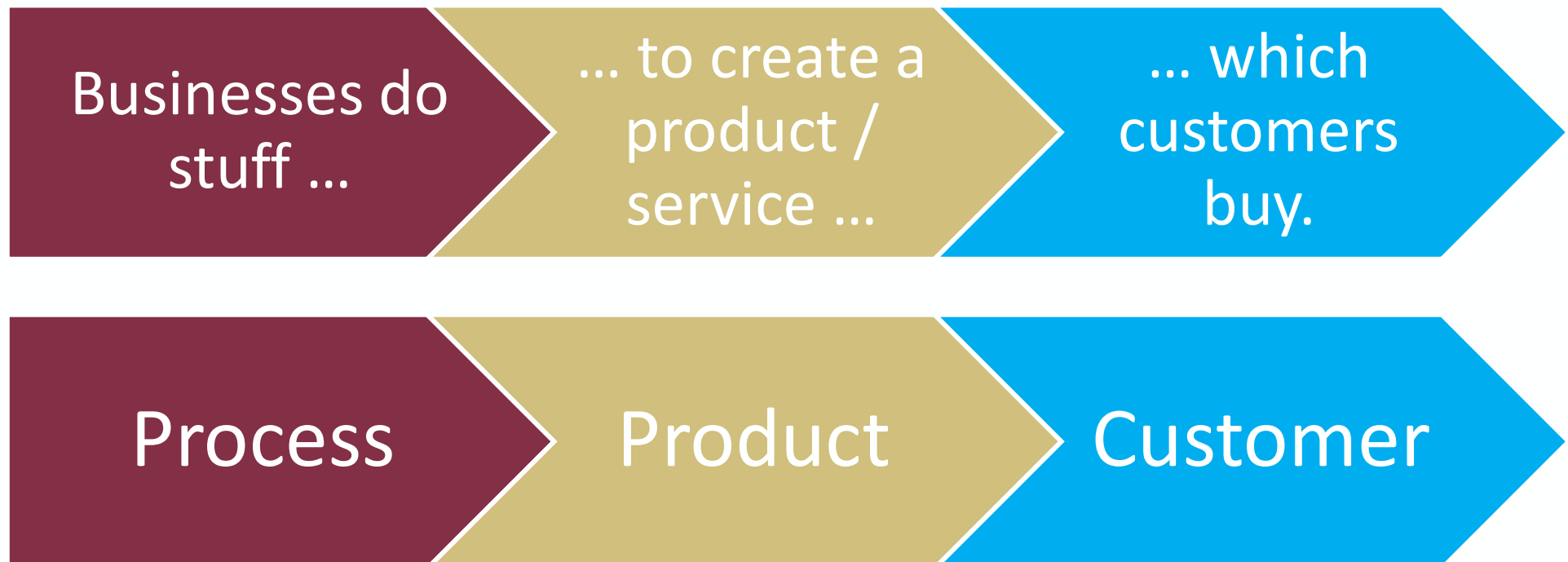
- i. have regard to reasonably foreseeable effects on the environment from those activities; and
- ii. have regard to the need for sustainable management of the environment. In this rule, sustainable management means management that meets the needs of the present without compromising the ability of future generations (including at least the future generations within the anticipated lifetime of the end products and by-products of those activities) to meet their own reasonably foreseeable needs.

How all businesses work...



Reference: Parrino, Yong, A., Dempsey, Morkel-Kingsbury, Ekanayake, James, Murray. (2019). *Fundamentals of Corporate Finance* (3rd ed.). Wiley

How all businesses work...



Something missing at all?

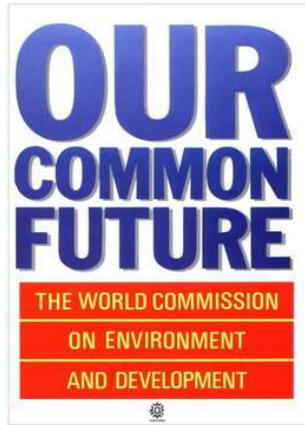
An Opportunity for Engineering

- Recognise and better manage risk
- Improve efficiency
- Revenue potential, growth
- Win more work → Tender responses, supply agreements etc. with a focus on sustainability
- Value alignment; attraction and retention of staff

What is sustainable development?

“Sustainable development is development that meets
the needs of the present generation
without compromising the ability
of future generations to meet
their needs”

- 1987 Brundtland Report of the World Commission on Environment and Development



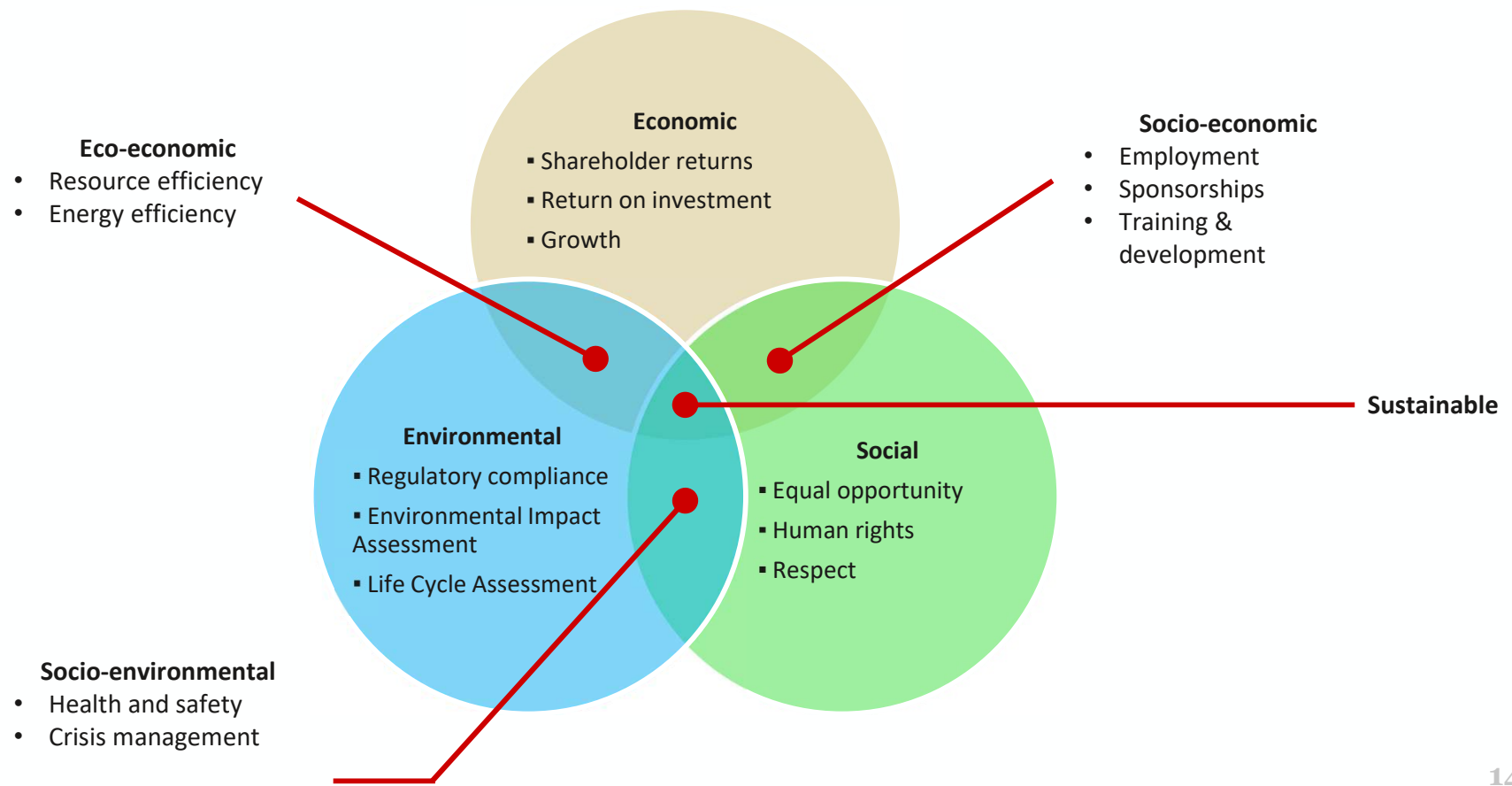
What is sustainable development?

Mō tātou, ā, mō kā uri ā muri ake nei.

For us and our children after us.

- Ngāi Tahu

Triple Bottom Line

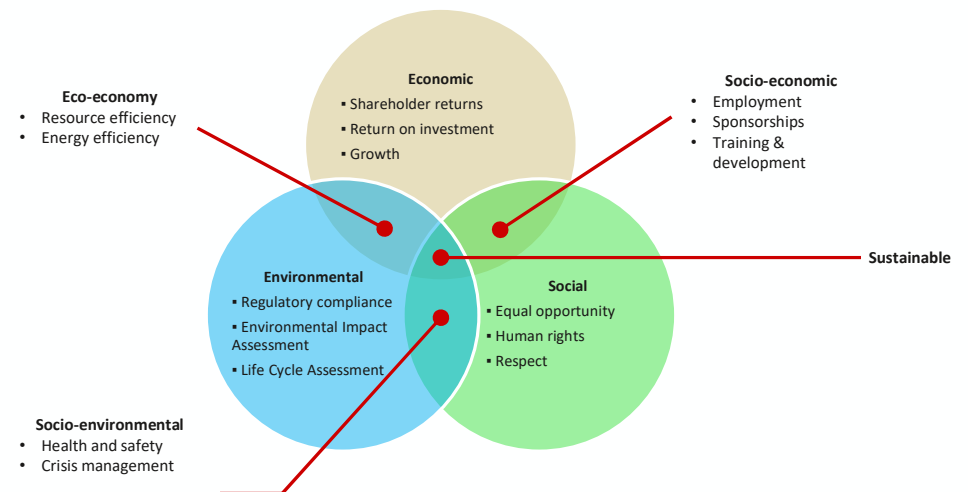


Triple Bottom Line

Often linked with decision making techniques

Eg Multi criteria decision analysis

- Identify factors of interest (decision criteria)
- Identify options
- Determine factor weightings
- Quantify factors for each option
- Rank / normalise → % scores for each option
- Score: weighting% x ranking%
- Highest score wins



Triple Bottom Line

1	Economic Indicators	
	Purchase Cost (\$ A)	19,000
	Average Fuel Efficiency (L/100 km)	1.11
	Unit Fuel Cost (cents/L)	161.5
	Undiscounted Lifetime Fuel Cost (\$ A)	359
	Monthly Average Depreciation Rate (%)	.92
2	Environmental Indicators	
	Lifetime Fuel Consumption (L)	222
	Carbon dioxide Emissions (CO2) kg	735.044
	Nitrogen Oxides Emissions (NOX) kg	03.484
	Sulfur dioxide (SO2) kg	02.336
3	Social Indicators	
	ANCAP Safety Rating (1 star lowest to 5 stars highest)	5
	Is there A Local Manufacturing Company ?	Australian Badged
Optional	Streamlined Whole of Life Costs (Purchase + Registration + Insurance + Operational - Resale)	23,628



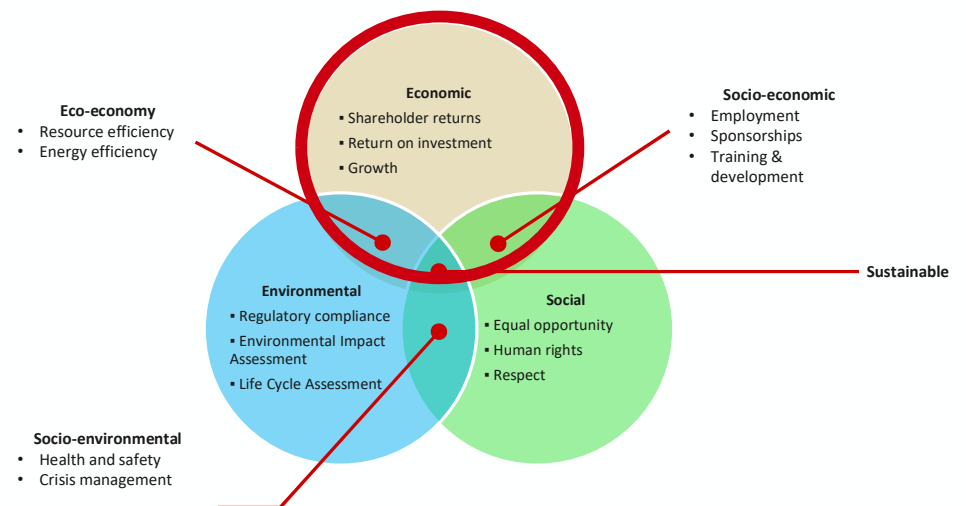
Example

Values	Egalitarian	State Govt. Preferred	Economist Preferred	Environmentally Preferred
Vehicle Cost	4.33	6.49	4.33	2.16
Fuel Cost	8.54	12.81	8.54	4.27
Depreciation Rate	3.74	3.74	7.47	1.87
Fuel Consumption	8.89	8.89	4.45	26.67
Carbon di-oxide	8.80	8.80	4.40	35.20
Nitrogen Oxides	8.79	8.79	4.39	26.36
Sulfur dioxide	8.47	8.47	4.23	25.40
Design for Safety	50.00	200.00	50.00	50.00
Local Mfg. Company	10	20	10	5
	111.5	278.0	97.8	176.9

Triple Bottom Line

Often linked with decision making techniques

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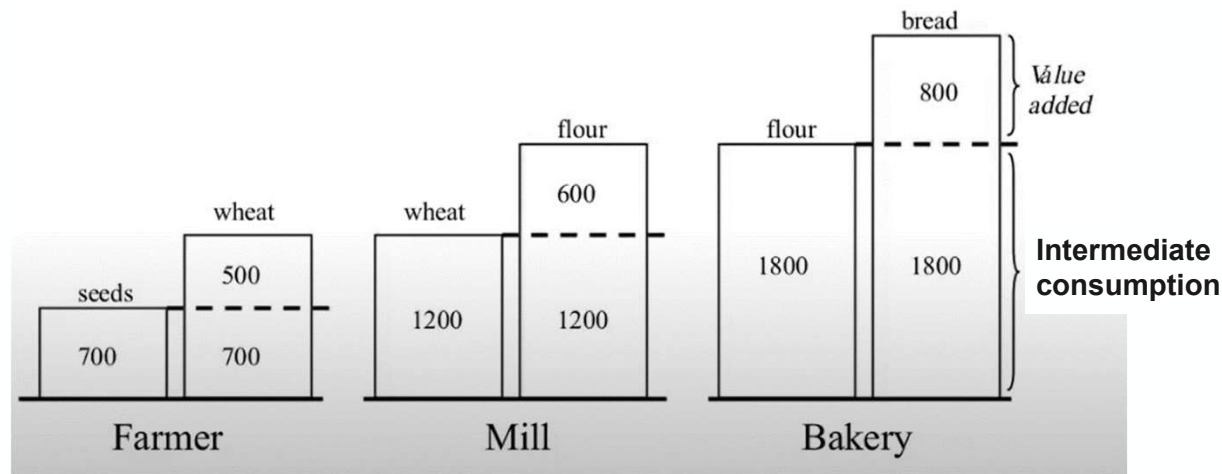


Economic assessment metrics

GDP

GDP: The value of the final goods and services produced in a country for a given time period

- Production-based:
 - Gross value added from production
 - To avoid double-counting, must subtract costs of inbound goods/services from value



Economic assessment metrics

- Net-present value techniques, inc. IRR, payback period
 - Typically deals with project costs:
 - Design
 - Initial investment (e.g. plant, equipment)
 - Ongoing costs (e.g. labour, overheads)
 - Ongoing income
 - Recovery costs (e.g. closure)
 - Recovery income (e.g. selling old equipment)
 - Questions:
 - Is it worth investing in?
 - How much money will the project make?
 - When do we get our money back?

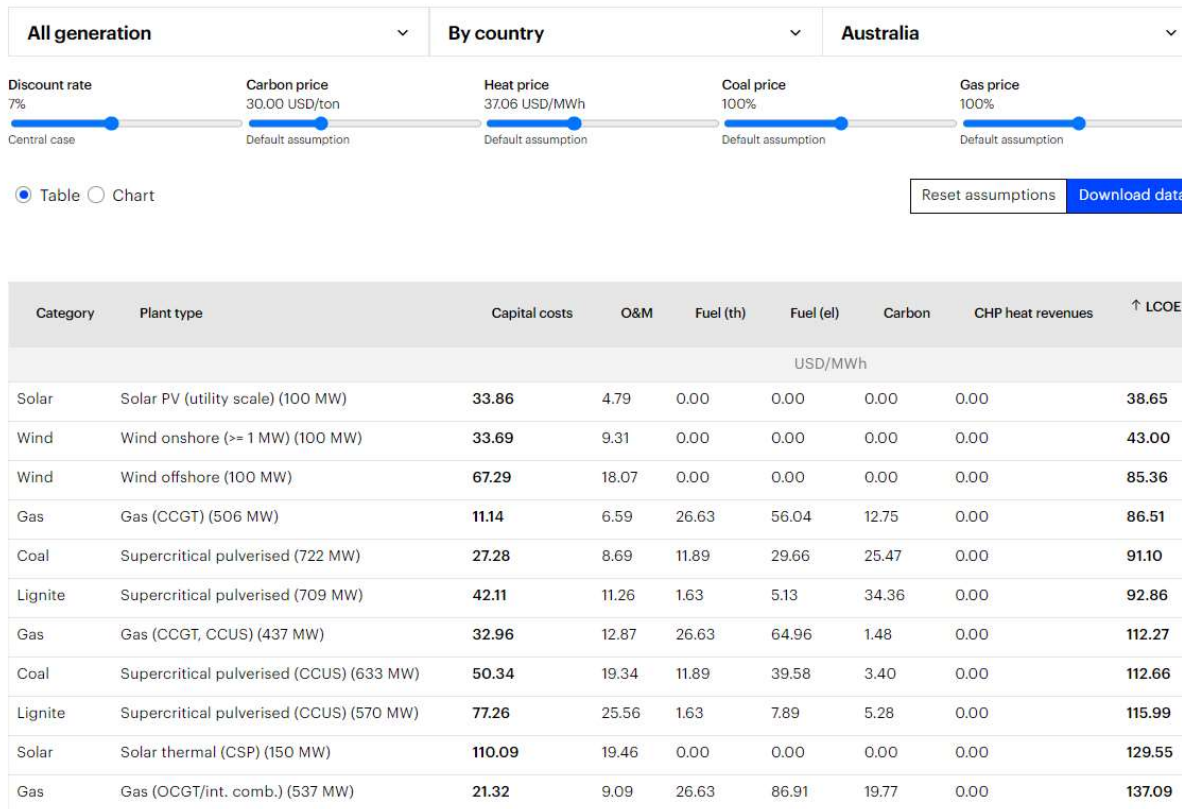
Economic assessment metrics

Levelised cost of energy (LCOE)

- Net present value per energy unit delivered
- Accounts for:
 - Expenditure (infrastructure/capital costs, fuel, financing, maintenance)
 - Tax credits (e.g. depreciation, renewable certificates)
- Underlying assumptions:
 - Capacity factor (actual / theoretical output)
 - Inflation rate(s) for income / expenses
 - Discount rate
 - Asset lifetime
- Typically quantified in \$ / MWh or c / kWh
- Price > LCOE → worth the investment!

Economic assessment metrics

Levelised cost of energy (LCOE)



<https://www.iea.org/data-and-statistics/data-tools/levelised-cost-of-electricity-calculator>

Economic assessment metrics

Levelised cost of energy (LCOE)

If nuclear is economically so cheap, why don't we?

→ Social concerns, risk, uncertainty and environmental impacts not well accounted for in LCOE (e.g. GHG abatement)



Economic assessment metrics

- Direct benefits, typically measured in monetary terms
 - At local level:
 - Voluntary investment into the community (by 3rd party)
 - Employment, including wage conditions
 - Retention of profits by community
 - Reduction in need to receive financial aid
 - At national level
 - Employment
 - Payments (e.g. tax)
 - Financial exchange with other industries
 - Reduction in need to give financial aid

Economic assessment metrics

- Indirect benefits, not typically/readily measured in monetary terms
 - At local level:
 - Suppliers of materials
 - Health and education
 - Increased productivity (outputs / intermediate inputs)
 - At national level
 - Infrastructure
 - Health

Economic assessment metrics

Integrating indirect impacts/benefits

- $\text{Cost} < \text{Price} < \text{Value}$
- Price and value does not always capture all costs
- Costs apportioned to a wide range of people not involved in a transaction are called **externalities**
 - Sometimes referred to as “monetised benefits”
- Negative **externalities** → Pollution, smoking
- Positive **externalities** → [Paid parental leave](#) (boosts GDP beyond what it costs)
- Other examples?

Economic assessment metrics

Integrating indirect impacts/benefits

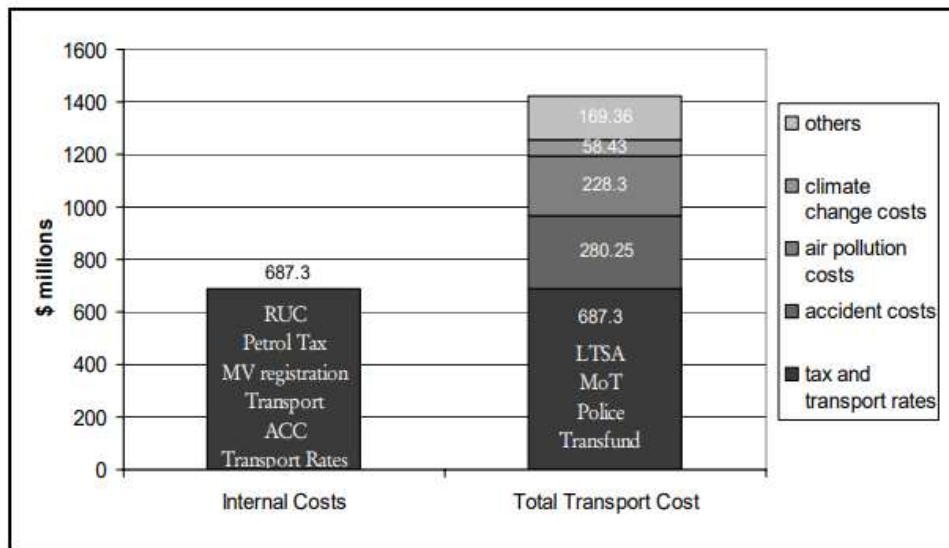


Figure 6: The total transport cost (external and internal) of public and private transport in the Auckland region in 2001 (MoT, 2002b). "Others" includes: external cost of noise, parking, land use change, water and soil pollution, resource use, barrier effects, waste and equity (Litman, 2002).

Source: Jacob, A., Craig, J.L. & Fisher, G. (2006). Transport cost analysis: A case study of the total costs of private and public transport in Auckland. *Environmental Science and Policy*, 9, 55-56. <https://doi.org/10.1016/j.envsci.2005.09.001>

- A more recent estimate of externalities for Auckland's transport system is \$2.2 - \$3.2 billion!

Source: Nunns, P. (2016, February 12). *Transport externalities in Auckland*, Greater Auckland.

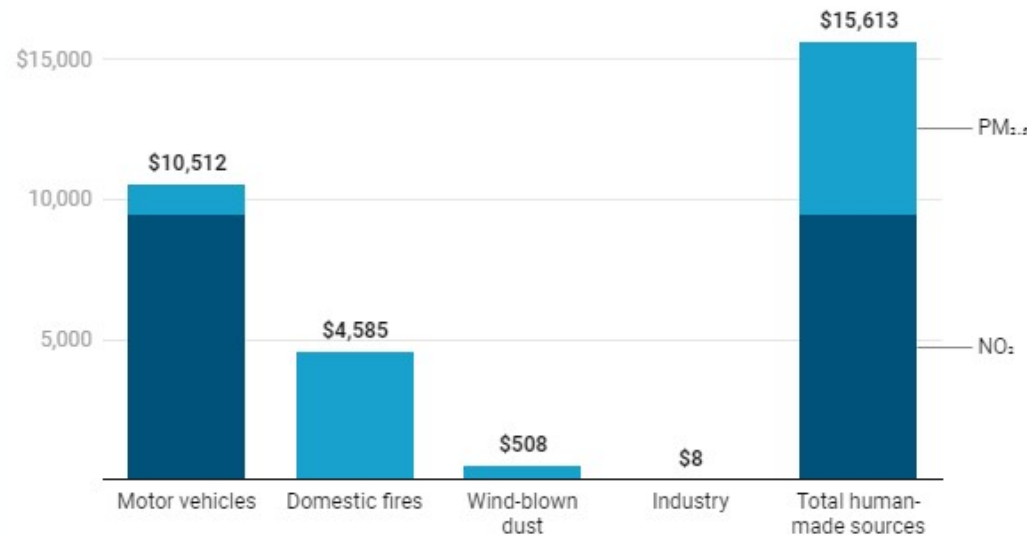
<https://www.greeterauckland.org.nz/2016/02/12/transport-externalities-in-auckland/>

Note that this source was not peer-reviewed

Economic assessment metrics

Integrating indirect impacts/benefits

- <https://www.ehinz.ac.nz/projects/hapinz3/>



Social costs from human made air pollution (NO₂ and PM_{2.5}) in New Zealand, by air pollutant and source of air pollution, 2016 (\$millions)

Economic assessment metrics

Economic management of externalities

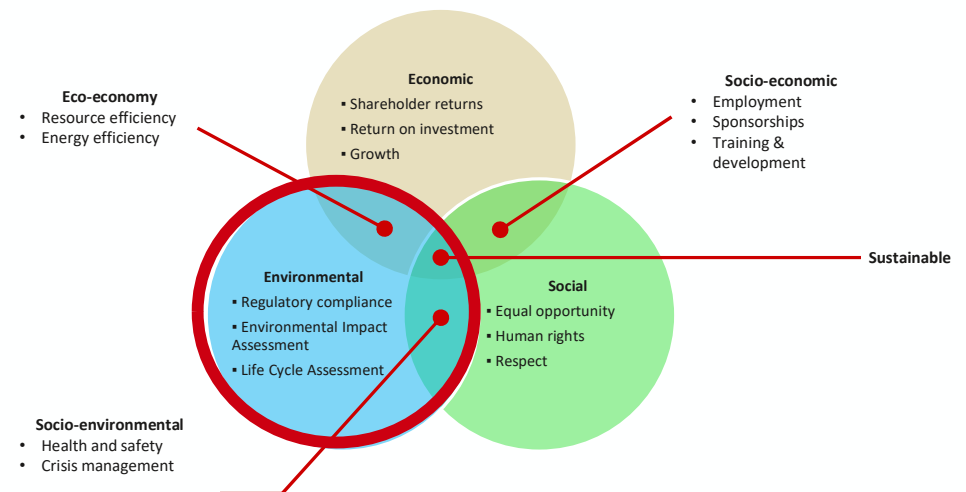
- Increased taxation (Pigouvian tax)
 - Covers the total cost of the externality
 - Government then uses to remediate / fix the damage caused
- Reduced taxation
 - Price signal to the market → makes the less damaging option more appealing
- Restrict supply (demand management) through regulations
- Tradeable permits

- Problems with these approaches?

Triple Bottom Line

Often linked with decision making techniques

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Environmental analysis

- Technique(s) used to collect information and the environmental effects of a project, which are used to inform a decision
 - Qualitative (descriptions)
 - Semi quantitative (descriptions matched to numerical values)
 - Quantitative (calculated numerical values)

Environmental analysis

There are statutory requirements that, by law, cannot be exceeded

- Typically these are dictated by legislation, acts of parliament, regulations or standards. Compliance is required. Otherwise government can act.
- Eg - levels of NO₂ and SO₂ according to the National Environmental Standards of Air Quality:
 - The current limits for NO₂ and SO₂ under the NES-AQ are:
 - 200 µg/m³ and 350 µg/m³ as a 1-hour average, with a maximum of 9 allowable exceedances per year

Consequences can include:

- Damage to human health / health of wildlife / flora and fauna species decline
- Prosecution with ensuing possibility of fines and imprisonment
- Reputational damage to companies

Environmental analysis

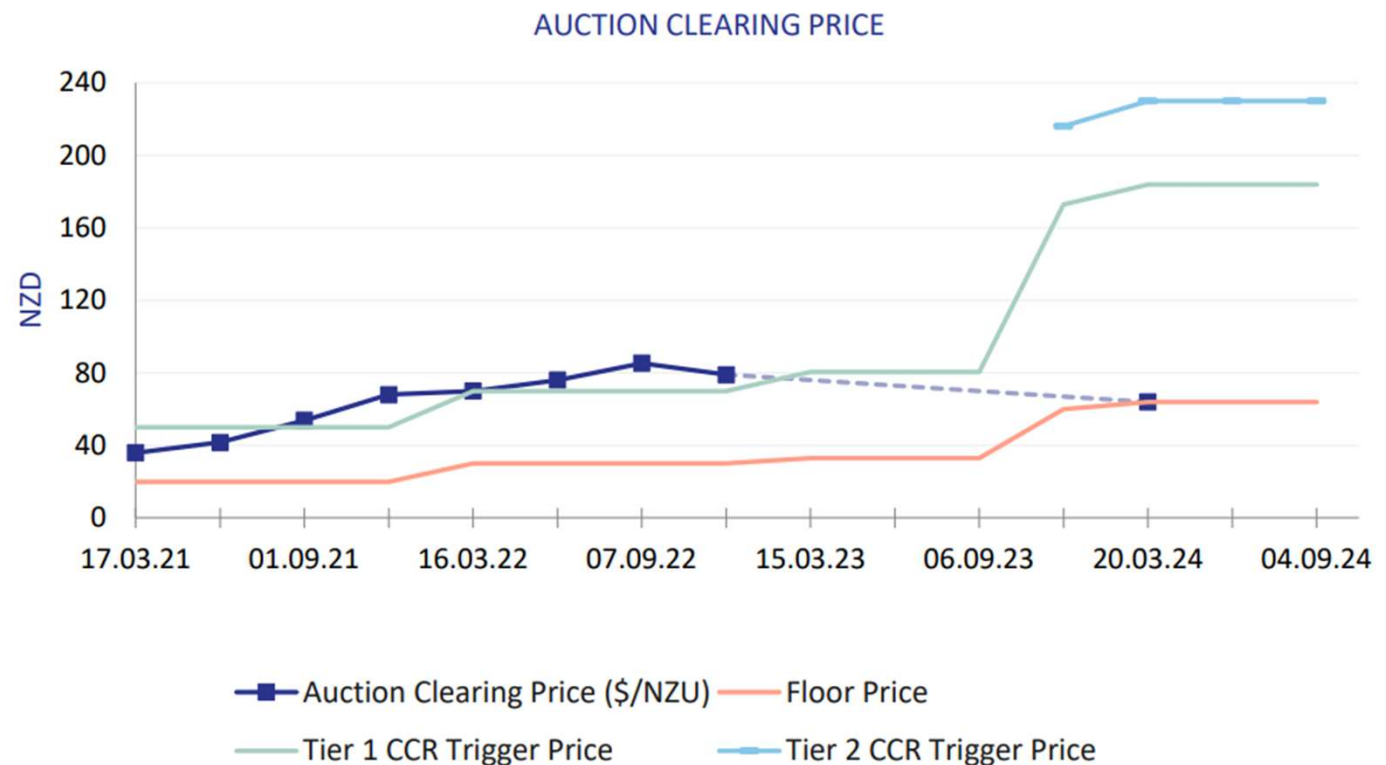
There are pollutants that can be emitted without penalty.

- For example, CO₂ emissions,
- CH₄ emissions

However, a company must participate in emissions trading if on schedule 3 and may participate if on schedule 4.

Hence, many companies must pay to emit.

Current floor price \$64 / t



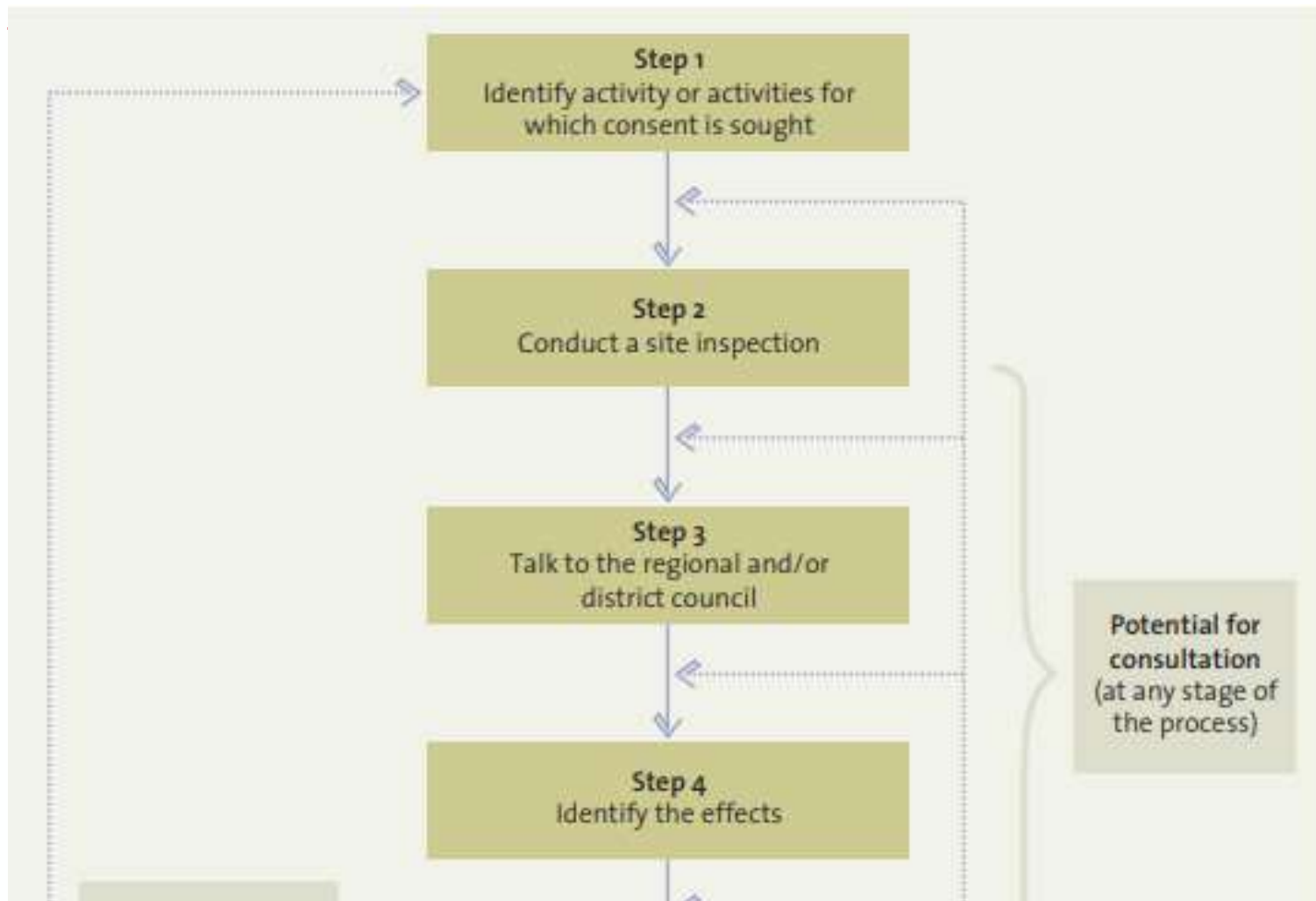
Environmental analysis

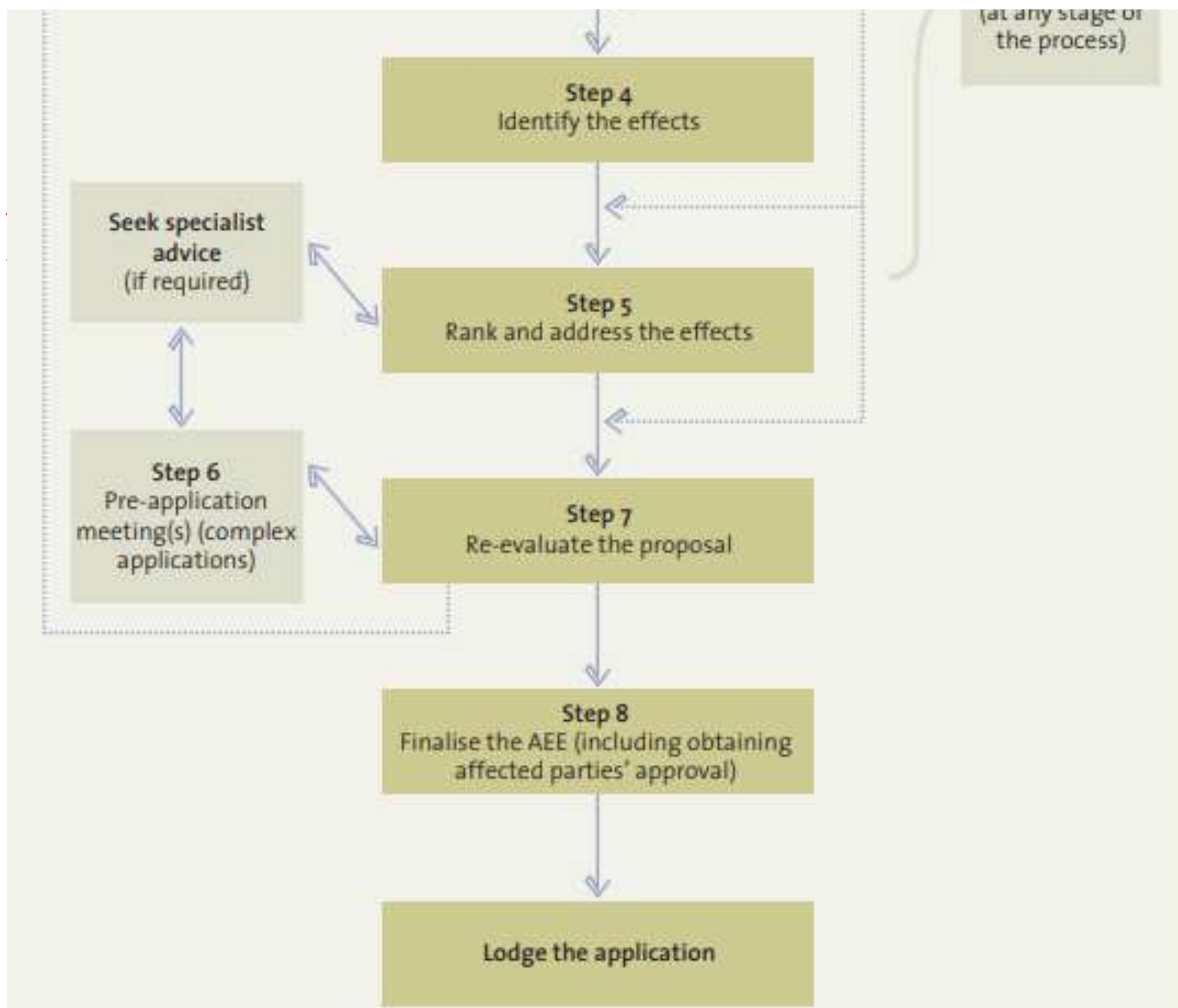
Assessment of Environmental Effects

- Forecasting environmental impacts and risks
- Describes existing environment
- Proposed project
- Development process
- Forecasts change
- Options to minimize:
 - once-off impacts
 - ongoing impacts → Environmental Management Plan
- Should be initiated during early planning, including critical stakeholders (e.g. government, local community, industry, iwi, hapū)

Environmental analysis

Assessment of Environmental Effects





Environmental analysis

Assessment of Environmental Effects

Impact identification matrix

- Identify impacts from interaction of project activities and specific environmental aspects
- Magnitude of importance, time frame
- Multiple options to highlight environmental benefits / impacts

Environmental aspect	Phase 1		Phase 2	
	Activity 1.1	Activity 1.2	Activity 2.1	Activity 2.2
Soil and geology				
Flora				
Fauna				
Noise				
Particulates				
Water quality				

Environmental analysis

Assessment of Environmental Effects

Impact identification matrix

- Identify impacts from interaction of project activities and specific environmental aspects
- Magnitude of importance, time frame
- Multiple options to highlight environmental benefits / impacts

Environmental aspect	Construction		Operation	
	Utilities	Buildings	Utilities	Buildings
Soil and geology	-	-		
Flora	-	-	-	++
Fauna	-	-		+
Noise	-	--	-	
Particulates	+			+
Water quality	+	-		+

Environmental analysis

Assessment of Environmental Effects

Impact identification matrix

- Identify impacts from interaction of project activities and specific environmental aspects
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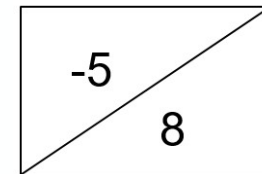
Environmental aspect	Construction		Operation	
	Utilities	Buildings	Utilities	Buildings
Soil and geology	-	-		
Flora	-	-	-	++
Fauna	-	-		+
Noise	-	--	-	
Particulates	+			+
Water quality	+	-		+

Environmental analysis

Assessment of Environmental Effects

Leopold Matrix

- Magnitude: - 10 (severe negative) → + 10 (highly positive)
- Importance: 1 lowest, 10 highest
- Judgement and stakeholder input

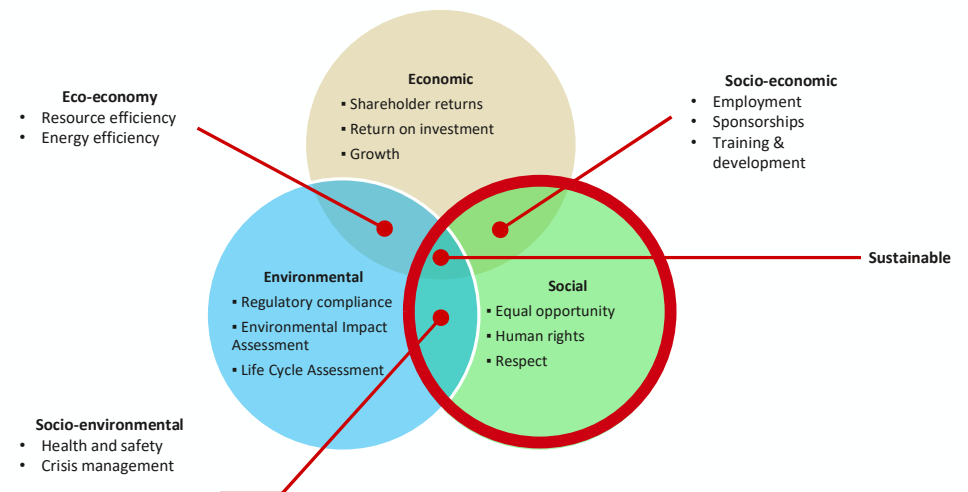


Environmental aspect	Construction		Operation	
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Soil and geology				
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Social impact assessment

- Analysing, monitoring and managing social consequences
 - Intended and unintended
 - Positive and negative
- Way of life
 - Live, work, play, social interactions
- Culture
 - Beliefs (inc. religious), customs, values, languages/dialects
- Community
 - Cohesion, stability, character, services, facilities
- Political systems
 - Participation, level of democratisation

Social impact assessment

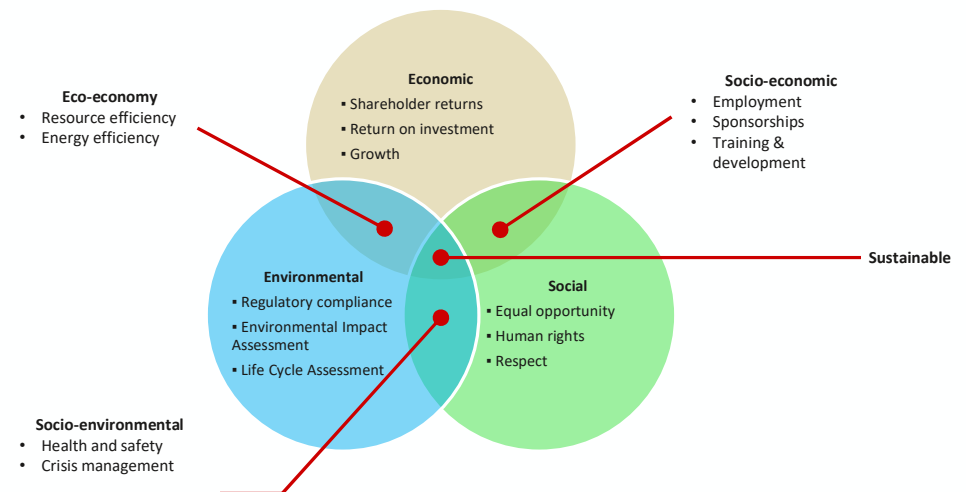
- Analysing, monitoring and managing social consequences
 - Intended and unintended
 - Positive and negative
- Well-being
 - Physical, mental, social wellbeing
- Rights
- Aspirations and fears
 - For the future, for self and others (e.g. children)

Usually measured in projects by increase in employment / safety of workers / community safety

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Multi-criteria decision modelling (Monday 7 Oct 3pm)

Triple Bottom Line

- Quantification / ranking challenges
- Trade-offs difficult to manage
 - Environmental vs societal vs financial vs ethical → values based assessment
 - Focus on improvements rather than trade-offs
- Criteria can be skewed to favour a preferred outcome
 - Problematic/controversial areas not addressed

In Summary

A triple bottom line analysis is a sustainability framework that considers the economic, social and environmental impacts of a project or business.

Ways to find indicators - Sustainable Development Goals



- A universal approach to sustainable development
- No distinction between developed and developing
- Goals
 - Targets
 - Indicators

Sustainable Development Goals



- Goal 9:

Build resilient infrastructure, promote inclusive and sustainable industrialisation and foster innovation

- Target 9.4:

By 2030, upgrade infrastructure and retrofit industries to make them sustainable, with increased resource-use efficiency and greater adoption of clean and environmentally sound technologies and industrial processes, with all countries taking action in accordance with their respective capabilities

- Indicator 9.4.1

CO₂ emission per unit of value added

Another option

The Mauri Model Decision Making Framework

- Towards decision making balance with regard to our social, economic, environmental and cultural well-being
- Developed by Dr Kipa Morgan (formerly UoA)

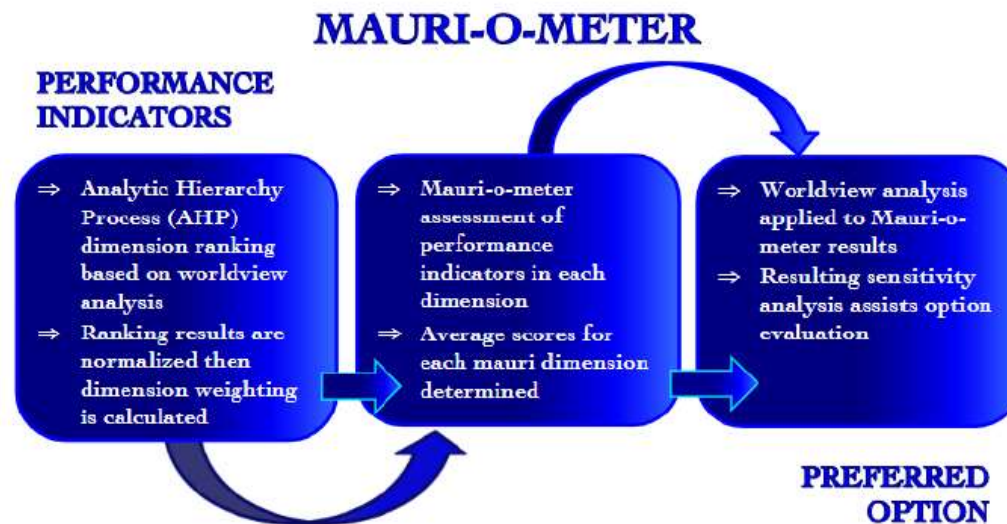


Figure 5 - Mauri Model methodology - assessment yields preferred option (Morgan 2008)

Another option

The Mauri Model Decision Making Framework

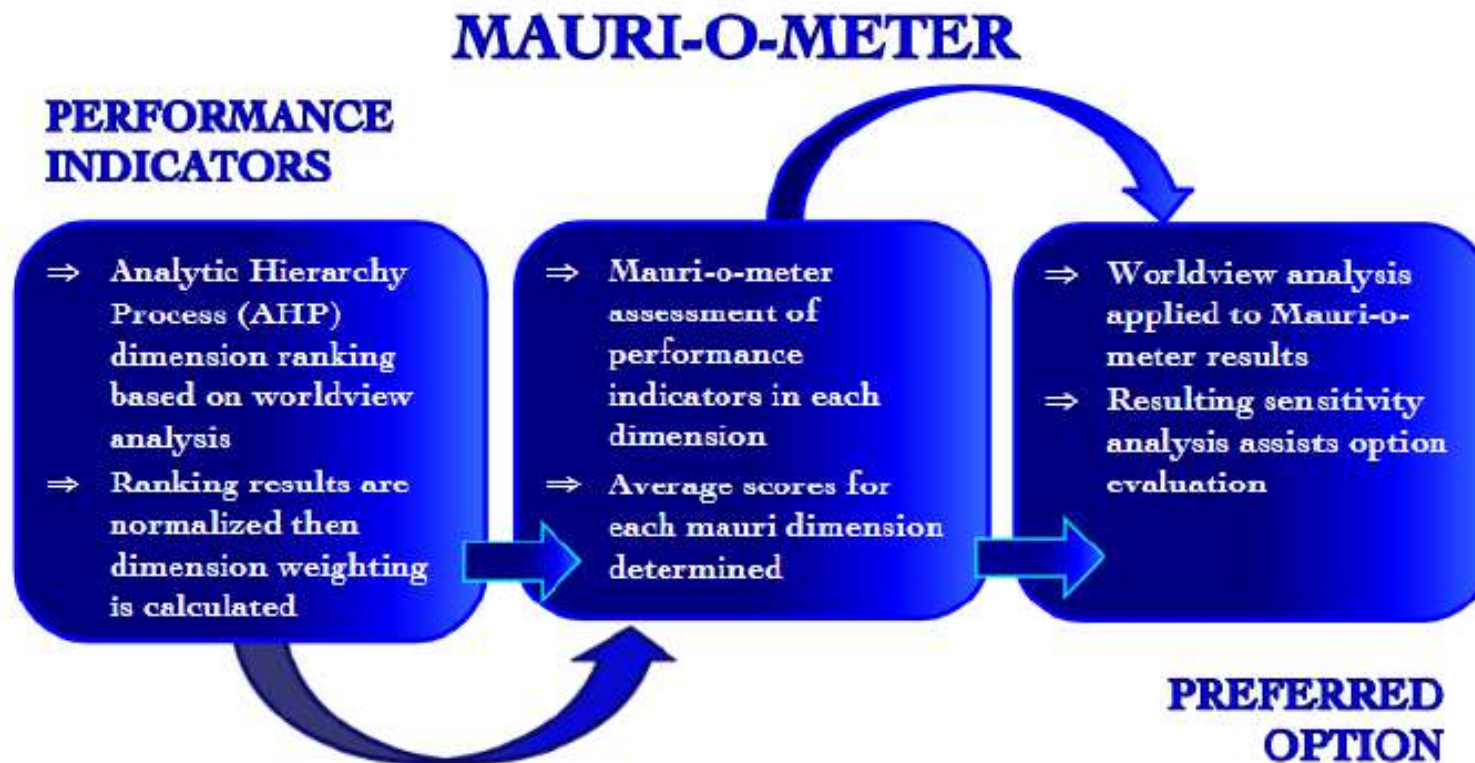


Figure 5 - Mauri Model methodology - assessment yields preferred option (Morgan 2008)

Another option

The Mauri Model Decision Making Framework

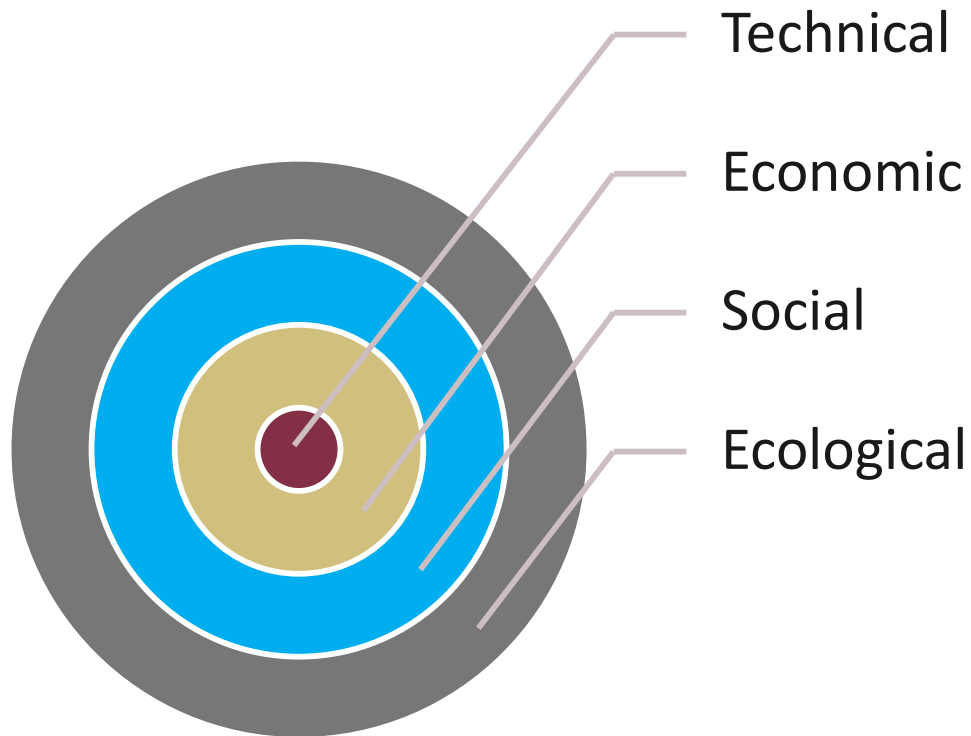
Advantages

- Provides a standardized framework for weighing up competing impacts
- reduced impact of personal constructs

Limitations

- Subjectivity of indicators
 - Context critical
 - Consensus needed
- emphasis on ethnographic dimensions
 - technical and economic considerations only in context of whanau
- contradictions of different world views

Another option Social – ecological system



... an “integrated complex system that includes social (human) and ecological (nature) subsystems in a two-way feedback relationship”
(Berkes 2011)

What's next?

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References

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- Mauri model: <https://www.maurimodel.nz/about-1>